Integer-factoring mechanisms for index-calculus work

Transfer assessment · 2026-11-01 · no performance or security promotion

1. Result at this scope

The useful import is a **soundness contract**: bind a residual's complete canonical identity, ambient curve and factor-base context, and action unit; eliminate only compatible equal-key records; then replay every merged row as an independent source-group equation before matrix insertion.

The host repository already has residual and one-/two-large-prime collectors, so no second collector was copied. Prime-cofactor saturation is rejected as a direct transfer because its `Y`-rough, `R <= Y^2` proof concerns prime divisors and arithmetic roots, not elliptic residual points. Moments, pole/Cauchy circuits and crossing profiles remain conditional leads.

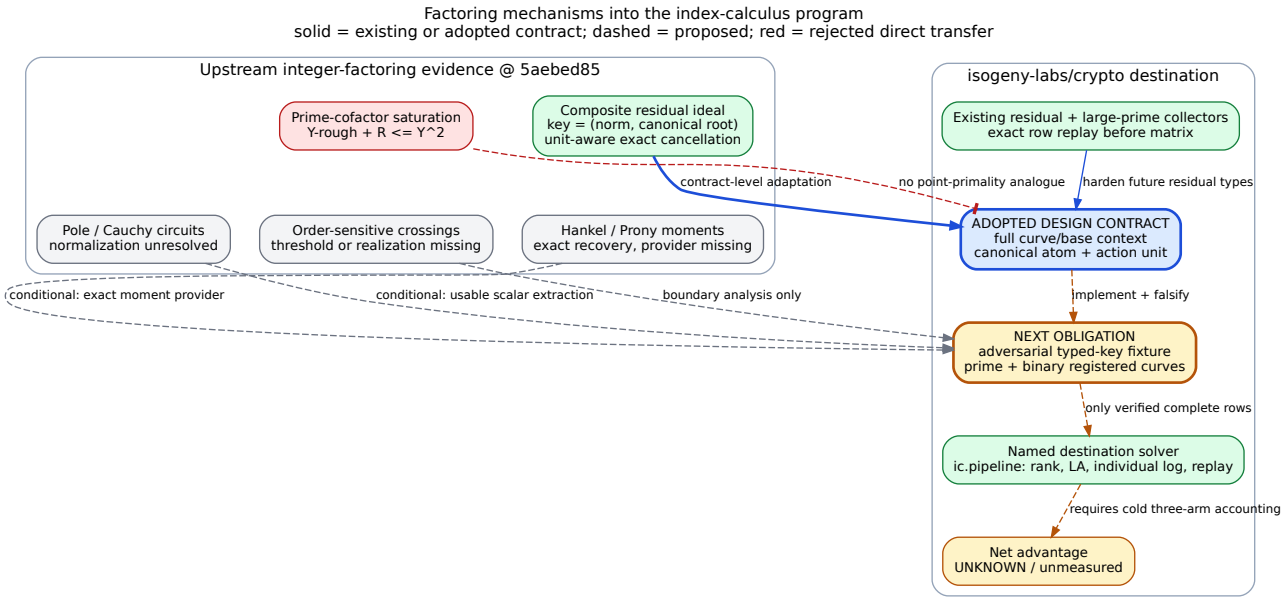


Figure 1: Typed transfer graph. Green nodes already exist, blue is the adopted contract, amber is open, and red is a rejected direct port.

2. Source and destination

The upstream source is `1337isnot1337/integer-factoring@5aebed85`. Its composite selector represents a partial as a fixed-base word times an oriented residual ideal keyed by `(norm, canonical root)`. Matching only the norm is unsound. Equal complete keys eliminate a temporary generator with a unit coefficient, and the result is checked as an exact ideal identity.

The destination is the existing `ic.pipeline` relation seam in `isogeny-labs/crypto@5b1a4826`: residual collection, signed Frobenius-orbit large-prime elimination, row verification, rank modulo `r`, linear algebra, individual-log recovery and independent scalar replay. There is no homomorphism between the factoring and elliptic objects; the transfer is an attributed algorithmic design pattern.

The future IC key must bind the full curve UID, subgroup order, factor-base and candidate identities, collector family, canonical residual atom, orbit/sign action, its unit modulo `r`, arity and infinity convention. A hash may index a table but cannot replace payload equality.

3. Evidence boundary

The upstream finite controls reduced charged leading-norm queries, while explicitly declining to treat those ratios as wall-clock gains.

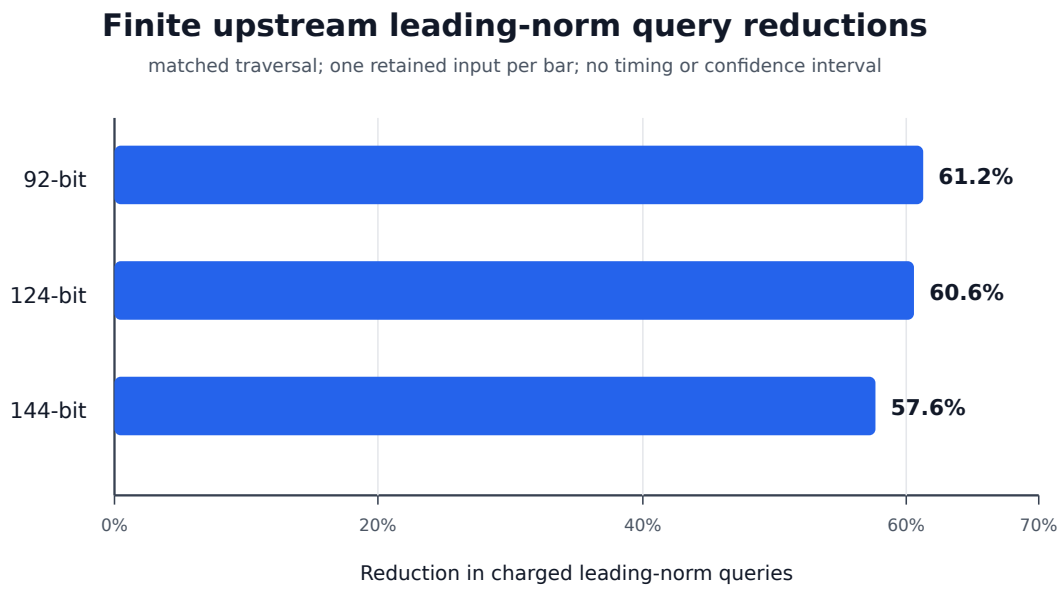


Figure 2: Recorded upstream query-count reductions. Population: one retained matched input at each displayed bit size; no timing sample or interval exists.

Input	Smooth-only queries	Collision queries	Reduction
92-bit	227,593	88,234	61.2%
124-bit	3,553,722	1,400,316	60.6%
144-bit	8,120,765	3,440,025	57.6%

The host controls are stricter for ECDLP interpretation:

- residual walks retained the birthday collision count even when residual updates became cheaper;
- the prime residual collector matched a curve-free carrier and remained about 3.8–4.9x counted rho at its cheapest fixed bases;
- the Koblitz double-large-prime graph showed a 2.1x finite constant over an equal-table control on one 39-bit rung, but its best cell remained 17.7x counted rho and no exponent was fitted; and
- multiplier bases already test the closest safe analogue of selective promotion under held-out complete-cold protocols.

These are finite, method-specific results. They neither refute every typed residual design nor support a new ECDLP advantage.

4. Dispositions

Mechanism	State	Reason
Canonical residual identity	Adopt	Prevents coarse-key cancellation; compatible with existing replay gates.
Prime residual saturation	Rejected direct	No elliptic point-primality or split-prime certificate exists.
Selective promotion	Conditional	Must create explicit columns or known-weight multipliers and charge downstream work.
Hankel / Prony moments	Conditional	Exact recovery exists, but no cheaper uniform moment provider is supplied.
Pole / Cauchy circuits	Conditional	Compact evaluation lacks a usable normalized row-extraction interface.
Crossing profiles	Boundary only	Threshold or realizing-tensor obligations remain open.

5. Next obligation

Freeze one small registered prime curve and one small registered binary curve. Construct adversarial partials with the same x-coordinate and opposite sign, the same orbit representative and different action units, and identical residual bytes under different curve/base identities. The typed key must reject incompatible pairs and reproduce the legacy collector's verified row on every compatible pair. An intentionally under-keyed arm is the negative control.

Only after this A0/A1 fixture passes should a held-out cold three-arm comparison be frozen. It must charge table construction and probes, canonicalization, memory, duplicate filtering, rank, linear algebra, individual log, recovery and verification. The current verdict is **unknown with this next discriminating check**.

6. Reproduction and sources

The full mechanism crosswalk and limitations are in `README.md`. Exact upstream paths, commit, Apache-2.0 license digest and SHA-256 bindings are in `SOURCE_MANIFEST.json`. The machine-readable gate is `ASSESSMENT.json` and is validated by `tools/transfer-assessment` v0.2.0.

Primary local sources: `research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md`, `research/ic_prime_residual_lp_20261026/README.md`, `research/ic_techniques_20261012/double_large_prime/README.md`, and `research/generator_multiplier_e2e_20261101/README.md` at `5b1a4826`.

No experiment was run, no canonical scoreboard or graph gained a verified result, and no existing canonical graph requires regeneration.